

Testing for anions and cations

Testing for gases:

- Chlorine gas: Hold a damp litmus paper to the container; turns colorless.
 - Hydrogen gas: Hold a lightened splint to the container; the gas will burn with a squeaky pop.
 - Oxygen gas: Hold a glowing splint to the container; the splint will immediately burst to flames.
 - Ammonia gas: Hold a red damp litmus paper to the container; turns blue
 - Carbon dioxide: Bubble the gas in limewater; turns from colorless clear to milky white.
- $$\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O(l)}$$
- (aq) (g) (s)

Testing for cations: Flame test:

* First, clean a platinum/nichrome wire by dipping it into a concentrated HCl solution, then hold it in a hot Bunsen burner flame until there is no color change. Then moisten the wire by dipping it into HCl again, and dip it in the salt. Hold the wire in the clear part of the blue Bunsen burner flame, and observe the color change.

- Lithium: Red
- Calcium: Orange-red
- Sodium: Yellow
- Barium: Light green
- Copper: Blue-green
- Potassium: Lilac

Testing for cations: Chemical test:

- For ammonium: NH_4^+ : add dropwise sodium hydroxide and heat gently to drive out ammonia, and then test for the gas.

Ionic equation: $\text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{NH}_3(\text{g}) + \text{H}_2\text{O(l)}$

- For copper: Cu^{2+} : * Add dropwise NaOH, a pale blue precipitate forms that will not dissolve if you add excess NaOH.

* Add dropwise ammonia: NH_3 , a pale blue precipitate forms that dissolves when adding excess NH_3 to form a deep blue solution.

Ionic equation: $\text{Cu}^{2+}(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{Cu(OH)}_2(\text{s})$

- For iron(II): Fe^{2+} : Add dropwise NaOH or NH_3 , or excess, the result is the same: a pale green precipitate forms.

Ionic equation: $\text{Fe}^{2+}(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{Fe(OH)}_2(\text{s})$

- For iron(III): Fe^{3+} : Add dropwise NaOH or NH_3 , or excess, the result is the same: a red-brown precipitate forms.

Ionic equation: $\text{Fe}^{3+}(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{Fe(OH)}_3(\text{s})$

- For aluminum: Al^{3+} : * Add dropwise NaOH, a white precipitate forms, that will dissolve in excess to form a colorless solution

* Add dropwise NH_3 , a white precipitate forms that will **not** dissolve in excess.

Ionic equation: $\text{Al}^{3+}(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{Al(OH)}_3(\text{s})$

- For zinc: Zn^{2+} : * Add dropwise NaOH, a white precipitate forms, that will dissolve in excess to form a colorless solution

* Add dropwise NH_3 , a white precipitate forms that will dissolve in excess to form a colorless solution.

Ionic equation: $\text{Zn}^{2+}(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{Zn(OH)}_2(\text{s})$

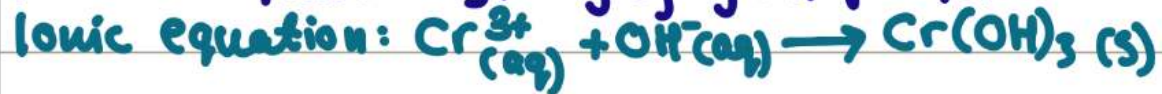
- For Calcium: Ca^{2+} : * Add dropwise NaOH , a white precipitate forms that will not dissolve in excess.

* Add dropwise NH_3 , a very slightly white precipitate will form, or no precipitate at all.



- For Chromium(III): Cr^{3+} : * Add dropwise NaOH , a green precipitate forms, that dissolves in excess to form a green solution.

* Add dropwise NH_3 , a grey-green precipitate forms that will not dissolve in excess.



Testing for anions:

* All nitrates are soluble

* All halides are soluble except for silver halides & lead halides.

* All sulfates are soluble except for calcium, barium, and lead sulfates.

* Only sodium, potassium and ammonium carbonates are soluble, all other carbonates are insoluble.

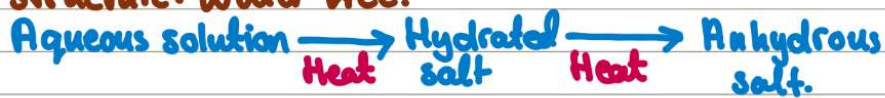
* All sodium, potassium, and ammonium salts are soluble.

→ Precipitation reaction: When two soluble salts / solutions react together to form an insoluble salt.

→ Double displacement reaction: When two compounds exchange ions to form new compounds.

Hydrated salt: A substance that contains water molecules as a part of its chemical structure.

Anhydrous salts: A substance that doesn't contain water molecules in its chemical structure: Water free.



Observation when having a hydrated salt:
Condensation at the mouth of the container.

Observations for hydrated copper sulfate: CuSO_4 :

* White fumes are given off

* The blue solid becomes white.

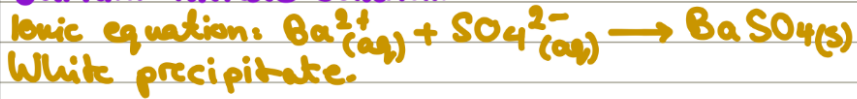
→ Testing for halides: Add dilute nitric acid for sterilization, then add aqueous silver nitrate solution, observe color change:

* Chloride: $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$
White precipitate.

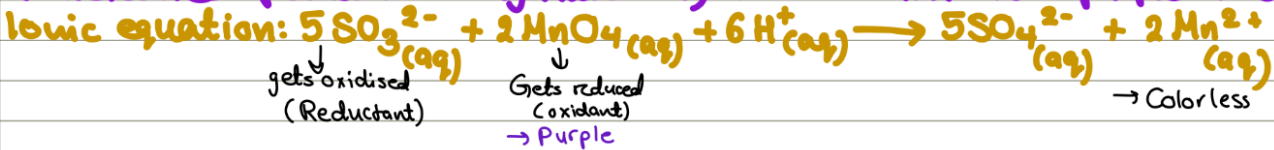
* Bromide: $\text{Ag}^+(\text{aq}) + \text{Br}^-(\text{aq}) \rightarrow \text{AgBr}(\text{s})$
Cream precipitate.

* Iodide: $\text{Ag}^+(\text{aq}) + \text{I}^-(\text{aq}) \rightarrow \text{AgI}(\text{s})$
Yellow precipitate.

→ Testing for sulfate ions: Add dilute hydrochloric acid, then add barium nitrate solution.



→ Testing for sulfite ions, and sulfur dioxide gas: Add a small amount of acidified potassium manganate(VII). It will turn from purple to colorless.



→ Testing for carbonate ions: Add dilute hydrochloric acid to the solution, then bubble the solution through limewater; turns from clear colorless to milky white.



→ Testing for nitrate ions: Add sodium hydroxide solution, then add some small pieces of aluminium foil, and heat gently, then test for ammonia gas by holding a red damp litmus paper to the container, that will turn blue.